

Location Recommender — Week 3 Update

CS335 Software Engineering · Microsoft Mentors: Dominic & Victor · 19 March 2026

Project Overview

A Letterboxd-style app for real-world places — bars, restaurants, and social spots. Users geo-tag locations, write reviews, and follow friends to see what they've visited nearby.

Confirmed Stack

- Frontend: React Native (Expo)
- Backend: Java + Spring Boot
- Database: PostgreSQL + PostGIS
- Auth: Azure Entra ID
- Infrastructure: Azure (Maps, Blob Storage, App Service)

MVP Features

- User authentication
- User profile
- Add & view locations (geo-tagged)
- Map view
- Write & read reviews
- Friend connections & friend feed
- Location ranking / score

Week 3 — Goals

- Get the project running locally for all team members
- Set up the database schema and have it running locally
- Build basic user authentication (sign up / login via Azure Entra ID)
- Establish folder structure for all areas of the project
- Produce a task list with complexity ratings for sprint planning

Completed This Week

Backend

- **Done** — Project structure set up (Spring Boot, JPA)
- **Done** — Local setup instructions written
- **Done** — REST endpoints for Users, Locations, Reviews
- **Done** — Azure Entra ID app registration initiated

Database

- **Done** — All 4 tables defined: users, locations, reviews, friendships
- **Done** — PostGIS extension and spatial index included in schema
- **Done** — Schema shared on GitHub

Frontend

- **Done** — Expo project scaffolded and pushed to GitHub
- **In progress** — App running on simulator
- **In progress** — Login screen (MSAL integration)

General

- **Done** — Repository restructured into clear folders
- **Done** — Task list with complexity ratings produced
- **Done** — Week-by-week roadmap confirmed

Still In Progress / To Do This Week

- **In progress** — PostgreSQL + PostGIS running locally on each team member's machine
- **In progress** — Login endpoint — Azure Entra ID token validation in Spring Boot
- **To do** — Frontend screen sketches (Login, Profile, Feed)
- **To do** — Frontend requirements list — what data each screen needs from the backend

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Task List & Roadmap · 19 March 2026

Task List — Complexity Ratings

E = Easy (a few hours) · **M** = Medium (half a day to a day) · **H** = Hard (multiple days, unknowns involved)

ID	Task	Difficulty	Week	Area
DB-1	Install PostgreSQL locally + enable PostGIS	M	3	Database
DB-2	Run the schema SQL (create all 4 tables)	E	3	Database
DB-3	Verify PostGIS geo queries (e.g. ST_Distance)	M	4	Database
DB-4	Add bio + profile_pic columns to users table	E	4	Database
DB-5	Add score column to locations table	E	6	Database
DB-6	Agree on schema versioning approach (SQL scripts in repo)	M	3	Database
BE-1	Fix DB connection (configure application.properties)	E	3	Backend
BE-2	Azure Entra ID — app registration on Azure portal	M	3	Backend
BE-3	Entra ID token validation in Spring Boot (SecurityConfig)	H	3	Backend
BE-4–6	CRUD endpoints — Users, Locations, Reviews	E	✓	Backend
BE-7–8	Filtered queries — reviews by location / by user	E	4	Backend
BE-9	Friend endpoints (send request, accept, list)	M	5	Backend
BE-10	Friend feed — reviews from friends	H	5	Backend
BE-11	Proximity search — locations within X km (PostGIS)	H	4	Backend
BE-12	Location score / ranking calculation	H	6	Backend
BE-13	Add bio + profile_pic to User model	E	4	Backend
FE-1–4	Expo running, folder structure, screen sketches, API requirements	E	3	Frontend
FE-5	Login screen + MSAL (Azure Entra ID)	H	3	Frontend
FE-6–9	Profile, Add location, Location detail, Write review screens	M	4	Frontend
FE-10	Map view screen (Azure Maps)	H	4	Frontend
FE-11–12	Friend feed screen, Friend request screen	M	5	Frontend

ID	Task	Difficulty	Week	Area
INT-1	Connect frontend to backend API	M	4	Integration
INT-2	End-to-end test: create user → add location → write review	M	4	Integration

Roadmap

Week 3 Mar 19	Setup, Scaffolding & Login PostgreSQL local setup · run schema · REST endpoints · Azure Entra ID registration · login endpoint · Expo running · login screen
Week 4 Mar 26	Locations, Reviews & Map Verify PostGIS · filtered review queries · proximity search · profile/location/review screens · map view (Azure Maps) · frontend connected to API
Week 5 Apr 2	Social Layer Friend request endpoints · friend feed endpoint · friend screens · photo uploads (if time allows)
Week 6 Apr 9	Polish & Testing Location ranking/score · end-to-end walkthrough · bug fixes · confirm Visit 1 date
Week 7 Apr 16	Presentation Prep (Visit 1) Slide deck · rehearse live demo · begin report outline
Week 8 Apr 23	Report & Submission Complete report · clean up GitHub · final stable build · submit April 27